

Linux commands

- cd
 - cd ~ - change working directory to home directory
 - cd - - change working directory to old working directory
 - cd .. - change working directory to parent directory
- ls
 - ls - list the contents of present working directory
 - ls path - list the contents of given path
 - ls -l - list the contents in detail format
 - type and permissions
 - Types of files
 - Permissions of files
 - r - read, w - write, x - execute
 - (rwx)user/owner, (rwx)group, (rwx)others
 - link count
 - user/owner
 - group
 - size
 - timestamp
 - name
 - ls -a - list all contents along with hidden
 - ls -A - list all contents along with hidden except . and ..
 - ls -li - list contents with inode number
 - inode number is unique number given to every file
 - ls -s - list content with size (number of blocks)
 - ls -S - list content in descending order of their sizes
- touch
 - if file does not exist, empty file is created
 - if file exist, timestamp of that file is changed
- stat
 - stat file - display information of file
 - stat file1 file2 - display information of file1 and file2
 - stat -c "format" file - display file information in given format
- head
 - head file - display first 10 lines
 - head -5 file - display first 5 lines
- tail
 - tail file - display last 10 lines

- tail -4 file - display last 4 lines
- sort
 - sort file - sort the content by alphabetically
 - sort -n file - sort the content by their value
 - sort command do not modify file content
- uniq
 - uniq file - display contents uniquely (truncate duplicate)
 - truncate duplicate content if it is consecutive
- rev filepath
 - Print each line reversed.
 - File contents are not modified.
- tac filepath
 - Print all lines in reverse order. The first line printed at last, while last line printed first.
- stat path
 - Display info about file or directory.
- alias
 - alias list="ls -l"
 - list will be alias/nick name to ls -l
 - list will give output same as ls -l
- unalias
 - unalias list
- which
 - which command
 - display the location of command executable.
- whereis
 - whereis command
 - display the location of command executable and also manual page location.

Linux Admin

- In Linux, Admin is called as "super-user".
- Admin's login name is "root".
- Most of modern Linux, disable "root" login (for security).
- To execute commands with admin privileges use "sudo" (if approved by system admin).
 - cmd> sudo apt update

- `cmd> sudo apt install vim gcc python3 python3-pip`
- `cmd> sudo snap install --classic code`

Directory commands

- `pwd` -- print present working directory (current directory)
- `cd` -- change directory (syntax> `cd dirpath`)
- `ls` - list directory contents (syntax> `ls dirpath`)
- `mkdir` -- make directory (syntax> `mkdir dirpath`)
- `rmdir` -- remove empty directory (syntax> `rmdir dirpath`)
- `cd`
 - `cd ~` - change working directory to home directory
 - `cd -` - change working directory to old working directory
 - `cd ..` - change working directory to parent directory

File commands

- `cat`
 - `cat > filepath` <-- create new file
 - `cat filepath` <-- display file contents
- `rm`
 - `rm filepath` <-- delete given file
 - `rm -r dirpath` <-- delete dir with all contents
- `mv`
 - `mv filepath destdirpath` <-- move given file into given dest directory
 - `mv dirpath destdirpath` <-- move given dir into given dest directory
 - `mv oldname newname` <-- rename given file
- `cp`
 - `cp filename newfilename` <-- copy file with new name/path.
 - `cp filepath destdirpath` <-- copy file into given dest dir with same name.
 - `cp -r dirpath destdirpath` <-- copy file into given dest dir with same name.

Redirection

- for every command input is taken from terminal, output is printed on terminal and error is also printed on terminal
- Standard streams (by default for every process, three files are opened)
 - `stdin`
 - `stdout`
 - `stderr`
- There are three types of redirections
 - input redirection
 - input will be taken from file instead of `stdin`
 - to do input redirection '`<`' symbol is used
 - `command < file`
 - output redirection

- output will be written into file instead of stdout
- to do output redirection '>' or '>>' symbol is used
- command > file
 - older content of file will be over written
- command >> file
 - content will be appneded into file at the end
- error redirection
 - error will be written into file instead of stderr
 - to do output redirection '2>' or '2>>' symbol is used
 - command 2> file
 - older content of file will be over written
 - command 2>> file
 - content will be appneded into file at the end

Pipe

- Using pipe, we can redirect output of any command to the input of any other command.
- Two processes are connected using pipe operator (|).
- Two processes runs simultaneously and are automatically rescheduled as data flows between them.
- If you don't use pipes, you must use several steps to do single task.
- command1 | command2
 - output of command1 will be given as input to command 2
- E.g.
 - who | wc

Shell meta characters

- '*' - zero or more occurances of any character
- '?' - one occurrence of any character

Regular Expressions

- Find a pattern in text file(s).
- Regular expressions are patterns used to match character combinations in strings.
- A regular expression pattern is composed of simple characters, or a combination of simple and special characters e.g. /abc/, /ab*c/
- Pattern is given using regex wild-card characters.
 - Basic wild-card characters
 - \$ - find at the end of line.
 - ^ - find at the start of line.
 - [] - any single char in give range or set of chars
 - [^] - any single char not in give range or set of chars
 - . - any single character
 - * - zero or more occurrences of previous character
 - Extended wild-card characters
 - ? - zero or one occurrence of previous character

- ▪ ▪ one or more occurrences of previous character
- {n} - n occurrences of previous character
- {,n} - max n occurrences of previous character
- {m,} - min m occurrences of previous character
- {m,n} - min m and max n occurrences of previous character
- () - grouping (chars)
- (|) - find one of the group of characters

grep

- Regex commands
 - grep - GNU Regular Expression Parser - Basic wild-card
 - egrep - Extended Grep - Basic + Extended wild-card
 - fgrep - Fixed Grep - No wild-card
- Command syntax
 - grep "pattern" filepath
 - grep [options] "pattern" filepath
 - -c : count number of occurrences
 - -v : invert the find output
 - -i : case insensitive search
 - -w : search whole words only
 - -R : search recursively in a directory
 - -n : show line number.

adduser

- sudo adduser username

passwd

- sudo passwd username

deluser

- sudo deluser username
 - home directory is not removed
- sudo deluser -r username
- sudo deluser --remove-home username